Orbital Mechanics for Propulsion Engineers

The astrodynamics behind `mission_analysis`

a tutorial for engineers who know thermodynamics and dynamics
but have not met orbits

MECE 4305 – Mechanics & Thermodynamics of Propulsion, Rocket Module
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How to read this tutorial

You know Newton's laws, energy and momentum, ODEs and vector calculus, and you know what a rocket engine does: it produces a velocity increment Δv at a propellant cost set by the rocket equation. What you may not know is where the Δv numbers in a mission come from. This tutorial builds that knowledge from the two-body problem up to the point where you can read, trust and extend every function in the `mission_analysis` package (the `sfmission` tool used in Week 10 and HW 1).

Each chapter follows the same pattern: the physics, derived at the level a mechanical engineer expects; a *worked example* with numbers computed by the package itself (every number in this document is generated by `calc/make_tutorial_numbers.py`, so text and code cannot drift apart); a *code box* naming the functions that implement the result; a *pitfall* where experience says people go wrong; and a few exercises, with answers in Appendix B.

Chapter	Question it answers	Module
1	How does a body move around a central mass?	`kepler`
2	How much Δv does it take to change an orbit?	`transfers`
3	How do we leave one body for another?	`patched_conics`
4	Where are the planets on a given date?	`ephemeris`
5	What transfer connects two positions in a given time?	`lambert`, `porkchop`
6	Why is the Moon a special case?	`lunar`
7	What does it cost to reach orbit?	`launch`
8	How does an atmosphere save propellant?	`aerobraking`
9	How does Δv become propellant and stages?	`staging`, `budget`
10	Putting it together: a Mars orbiter	all

Units and symbols. Kilometres, seconds and kilograms; angles in radians in formulas and degrees in text. One meaning per symbol: μ is *only* the gravitational parameter GM; $\mathcal{E}$ is specific mechanical energy; ζ, σ_i, π_i, R_i are the mass fractions of the lecture notes (Chapter 9); ε is reserved for nozzle area ratio and never appears here. A full table is in Appendix A.

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Chapter 1

One body around another: the two-body problem

1.1 Setting up the problem

Two bodies, one equation. Place a planet of mass M and a spacecraft of mass m in an inertial frame at positions $\boldsymbol{R}_M$ and $\boldsymbol{R}_m$ (Fig. 1.1). Newton's law of gravitation gives each the force $GMm/|\boldsymbol{R}_m - \boldsymbol{R}_M|^2$ directed toward the other, and Newton's second law gives their accelerations:

$$M\,\ddot{\boldsymbol{R}}_M = +\frac{GMm}{|\boldsymbol{R}_m - \boldsymbol{R}_M|^3}\,(\boldsymbol{R}_m - \boldsymbol{R}_M) \qquad \text{(planet pulled toward the spacecraft)},$$

$$m\,\ddot{\boldsymbol{R}}_m = -\frac{GMm}{|\boldsymbol{R}_m - \boldsymbol{R}_M|^3}\,(\boldsymbol{R}_m - \boldsymbol{R}_M) \qquad \text{(spacecraft pulled toward the planet)}.$$

Two things follow by adding and by subtracting. *Adding* gives $M\ddot{\boldsymbol{R}}_M + m\ddot{\boldsymbol{R}}_m = \boldsymbol{0}$: the centre of mass $\boldsymbol{R}_{cm} = (M\boldsymbol{R}_M + m\boldsymbol{R}_m)/(M+m)$ moves in a straight line at constant speed, so nothing interesting happens to it and we may attach an inertial frame to it. *Dividing each equation by its mass and subtracting* removes the frame altogether: with the relative position $\boldsymbol{r} \equiv \boldsymbol{R}_m - \boldsymbol{R}_M$ (the spacecraft as seen from the planet) and $r = |\boldsymbol{r}|$,

$$\ddot{\boldsymbol{r}} = -\frac{Gm}{r^3}\boldsymbol{r} - \frac{GM}{r^3}\boldsymbol{r} = -\frac{G(M+m)}{r^3}\,\boldsymbol{r} \;\approx\; -\frac{\mu}{r^3}\,\boldsymbol{r}, \qquad \mu \equiv GM. \tag{1.1}$$

The approximation is $m \ll M$: for a 10 t spacecraft about Earth, $m/M = 1.7\times10^{-21}$, so the planet's own acceleration toward the spacecraft is immeasurable and the planet-centred frame is inertial for every practical purpose. When the two masses are comparable the exact form is used with $\mu \to G(M+m)$: the Moon's motion about Earth ($m/M = 0.0123$) is computed with $\mu_\oplus + \mu_{Moon}$, which is how the package obtains the Moon's mean motion.



Figure 1.1: Left: the two-body geometry; Eq. (1.1) describes the relative vector $\boldsymbol{r}$. Right: polar decomposition of the velocity in the orbital plane into radial $\dot{r}$ and transverse $r\dot{\nu}$ components; the flight-path angle ϕ is measured from the local horizontal.

What has been assumed. (i) Point masses—equivalently, spherically symmetric bodies, because by Newton's shell theorem a spherical shell attracts an external point as if its mass were at its centre. Earth's oblateness (J_2) adds an acceleration of about $1.5J_2(R/r)^2 \approx 0.15\%$ of the central term in LEO; it rotates the orbit plane slowly and is ignored here. (ii) No other forces: no third body, drag, or radiation pressure. The Sun's tidal pull on a spacecraft at the Moon's distance is about 1% of Earth's attraction, and at a few hundred thousand kilometres more it wins—that boundary is the sphere of influence of Chapter 3. (iii) Newtonian gravity; relativistic corrections are parts in 10^8. Within these assumptions Eq. (1.1) is exact, and it is the same equation as any inverse-square central force (the electrostatic Kepler problem).

Why μ and not G and M. What tracking stations measure is the acceleration of spacecraft, i.e. GM directly (and Kepler's third law, Eq. (1.10), gives $GM = 4\pi^2 a^3/T^2$ from any orbit). G by itself is one of the least precisely known constants of physics (2×10^{-5} relative), so M inherits that uncertainty while GM does not: $\mu_\oplus$ is known to ten digits. Always carry μ.

Body	μ [km^3/s^2]	mean radius [km]
Sun	1.3271×10^{11}	695,700
Earth	398600.4	6378.1
Moon	4902.8	1,737.4
Mars	42828.4	3,396.2
Jupiter (system)	1.2671×10^{8}	71,492

What kind of problem this is. Equation (1.1) is a second-order vector ODE: six state variables ($\boldsymbol{r}$, $\boldsymbol{v} = \dot{\boldsymbol{r}}$) and six initial conditions. One could integrate it numerically—and for perturbed problems one does—but the two-body problem has enough conserved quantities (*integrals of the motion*) that the whole trajectory can be written down in closed form. Finding them is the next section; the identities used are collected in the box below.

Vector identities used in this chapter

With $r = |\boldsymbol{r}|$, $\boldsymbol{v} = \dot{\boldsymbol{r}}$ and $\hat{\boldsymbol{r}} = \boldsymbol{r}/r$:

$$r^2 = \boldsymbol{r}\cdot\boldsymbol{r} \;\Rightarrow\; 2r\dot r = 2\,\boldsymbol{r}\cdot\boldsymbol{v} \quad\Rightarrow\quad \boldsymbol{r}\cdot\boldsymbol{v} = r\dot r \qquad \text{(the radial component of } \boldsymbol{v} \text{ is } \dot r),$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{1}{r}\Big) = -\frac{\dot r}{r^2} = -\frac{\boldsymbol{r}\cdot\boldsymbol{v}}{r^3}, \qquad \frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\Big(\frac{\boldsymbol{r}}{r}\Big) = \frac{\boldsymbol{v}}{r} - \frac{\boldsymbol{r}\,\dot r}{r^2} = \frac{r^2\boldsymbol{v} - (\boldsymbol{r}\cdot\boldsymbol{v})\,\boldsymbol{r}}{r^3},$$

$$\boldsymbol{a}\times(\boldsymbol{b}\times\boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a}\cdot\boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a}\cdot\boldsymbol{b}), \qquad (\boldsymbol{a}\times\boldsymbol{b})\cdot\boldsymbol{c} = \boldsymbol{a}\cdot(\boldsymbol{b}\times\boldsymbol{c}), \qquad \frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{a}\times\boldsymbol{b}) = \dot{\boldsymbol{a}}\times\boldsymbol{b} + \boldsymbol{a}\times\dot{\boldsymbol{b}}.$$

1.2 What is conserved

Both conservation laws come from Eq. (1.1) by a single vector operation each: the cross product with $\boldsymbol{r}$ kills the right-hand side (the force is central), the dot product with $\boldsymbol{v}$ turns it into a total time derivative (the force is conservative).

Angular momentum. Take the cross product of $\boldsymbol{r}$ with Eq. (1.1):

$$\boldsymbol{r}\times\ddot{\boldsymbol{r}} = -\frac{\mu}{r^3}\,\boldsymbol{r}\times\boldsymbol{r} = \boldsymbol{0}.$$

The gravitational force acts along $\boldsymbol{r}$, so it exerts no torque about the planet. On the other hand, by the product rule,

$$\frac{\mathrm{d}}{\mathrm{d}t}(\boldsymbol{r}\times\boldsymbol{v}) = \dot{\boldsymbol{r}}\times\boldsymbol{v} + \boldsymbol{r}\times\dot{\boldsymbol{v}} = \boldsymbol{v}\times\boldsymbol{v} + \boldsymbol{r}\times\ddot{\boldsymbol{r}} = \boldsymbol{r}\times\ddot{\boldsymbol{r}},$$

and therefore

$$\boldsymbol{h} \equiv \boldsymbol{r}\times\boldsymbol{v} = \text{const.} \tag{1.2}$$

$\boldsymbol{h}$ is the *specific* angular momentum (per unit spacecraft mass, units km^2/s); its magnitude is $h = rv\sin\gamma$ with γ the angle between $\boldsymbol{r}$ and $\boldsymbol{v}$, i.e. r times the component of velocity perpendicular to the radius. Three consequences:

1. **The motion is planar.** $\boldsymbol{r}\cdot\boldsymbol{h} = \boldsymbol{r}\cdot(\boldsymbol{r}\times\boldsymbol{v}) = 0$ at every instant, so $\boldsymbol{r}$ always lies in the fixed plane through the planet's centre normal to $\boldsymbol{h}$. The six-dimensional problem has become two-dimensional; the orientation of the plane (i, Ω in Section 1.6) is fixed by the initial conditions and never changes.
2. **Kepler's second law.** In polar coordinates (r,ν) in that plane (Fig. 1.1, right) the velocity is $\boldsymbol{v} = \dot{r}\,\hat{\boldsymbol{r}} + r\dot{\nu}\,\hat{\boldsymbol{\nu}}$; only the transverse component contributes to the cross product, so

$$h = r^2\dot{\nu} = rv\cos\phi, \tag{1.3}$$

   where ϕ is the *flight-path angle* of the velocity above the local horizontal ($\cos\phi = \sin\gamma$). The area swept by the radius vector in $\mathrm{d}t$ is $\mathrm{d}A = \frac{1}{2}r\,(r\,\mathrm{d}\nu)$, so $\mathrm{d}A/\mathrm{d}t = h/2$ is constant: equal areas in equal times. A spacecraft therefore moves fast where r is small (periapsis) and slowly where it is large (apoapsis); integrated over a closed orbit this gives the period in Section 1.5.
3. **A geometric handle.** Because h is known from any one point of the orbit ($h = r_p v_p$ at periapsis, where $\phi = 0$), Eq. (1.3) gives the flight-path angle anywhere else: $\cos\phi = h/(rv)$. The lunar patched conic of Chapter 6 uses exactly this to find the arrival geometry at the Moon's sphere of influence.

Energy. Take the dot product of $\boldsymbol{v}$ with Eq. (1.1):

$$\boldsymbol{v}\cdot\dot{\boldsymbol{v}} = -\frac{\mu}{r^3}\,\boldsymbol{r}\cdot\boldsymbol{v}.$$

The left side is $\frac{1}{2}\,\mathrm{d}(\boldsymbol{v}\cdot\boldsymbol{v})/\mathrm{d}t = \mathrm{d}(v^2/2)/\mathrm{d}t$. For the right side use $\boldsymbol{r}\cdot\boldsymbol{v} = r\dot{r}$ from the box: $-\mu\dot{r}/r^2 = \mathrm{d}(\mu/r)/\mathrm{d}t$. Hence $\mathrm{d}(v^2/2 - \mu/r)/\mathrm{d}t = 0$ and

$$\mathcal{E} \equiv \frac{v^2}{2} - \frac{\mu}{r} = \text{const.} \tag{1.4}$$

This is conservation of mechanical energy per unit mass: kinetic energy $v^2/2$ plus the gravitational potential energy $-\mu/r$. The potential is the work per unit mass needed to carry the spacecraft from r to infinity against gravity, $\int_r^\infty \mu/r'^2\,\mathrm{d}r' = \mu/r$, with the zero of potential placed at infinity—the natural choice when the destination may be another planet. Near a planet's surface it reduces to the familiar gh: $-\mu/(R+h) \approx -\mu/R + (\mu/R^2)h = \text{const} + gh$ with $g = \mu/R^2$. Units are $\mathrm{km}^2/\mathrm{s}^2$, i.e. MJ/kg: a spacecraft in a 420-km orbit has $\mathcal{E} = -29.32\,\mathrm{MJ/kg}$, while at rest on the ground $\mathcal{E} = -\mu/R = -62.5\,\mathrm{MJ/kg}$, so reaching LEO costs about 33 MJ per kilogram—three times the 10–13 MJ released by burning a kilogram of LOX/kerosene or LOX/hydrogen propellant, which is why launch vehicles are mostly propellant.

The sign of $\mathcal{E}$ classifies the motion without solving anything:

- $\mathcal{E} < 0$: since $v^2/2 \geq 0$, the spacecraft can never exceed $r_{max} = -\mu/\mathcal{E}$; it is *bound* and, as the next section shows, moves on an ellipse whose size is set by $\mathcal{E}$ alone.
- $\mathcal{E} = 0$: it reaches $r \to \infty$ with $v \to 0$—the *parabolic*, or escape, case. At any radius its speed is $v_{esc} = \sqrt{2\mu/r}$.
- $\mathcal{E} > 0$: it reaches infinity with a residual speed $v_\infty = \sqrt{2\mathcal{E}}$—the *hyperbolic* case that every interplanetary departure and arrival is.

Energy knows the *size* of the orbit but nothing about its shape or orientation; angular momentum knows the shape (through $p = h^2/\mu$ below) and the plane. Together with the direction of periapsis they are the six constants the six-dimensional problem needs, and Eq. (1.4) with one velocity difference is behind most Δv estimates in this tutorial.

1.3 The orbit is a conic section

Cross Eq. (1.1) with $\boldsymbol{h}$ and use $\boldsymbol{r} \times (\boldsymbol{r} \times \boldsymbol{v}) = \boldsymbol{r}(\boldsymbol{r} \cdot \boldsymbol{v}) - \boldsymbol{v}r^2$:

$$\ddot{\boldsymbol{r}} \times \boldsymbol{h} = -\frac{\mu}{r^3}\left[\boldsymbol{r}(\boldsymbol{r} \cdot \boldsymbol{v}) - \boldsymbol{v}r^2\right] = \mu\left[\frac{\boldsymbol{v}}{r} - \frac{\boldsymbol{r}\,\dot{r}}{r^2}\right] = \mu\,\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\boldsymbol{r}}{r}\right).$$

Since $\boldsymbol{h}$ is constant the left side is $\mathrm{d}(\boldsymbol{v} \times \boldsymbol{h})/\mathrm{d}t$, so integrating, $\boldsymbol{v} \times \boldsymbol{h} = \mu(\hat{\boldsymbol{r}} + \boldsymbol{e})$ with a constant vector $\boldsymbol{e}$, the *eccentricity vector*. Dotting with $\boldsymbol{r}$ and using $\boldsymbol{r} \cdot (\boldsymbol{v} \times \boldsymbol{h}) = (\boldsymbol{r} \times \boldsymbol{v}) \cdot \boldsymbol{h} = h^2$:

$$r(\nu) = \frac{p}{1 + e\cos\nu}, \qquad p \equiv \frac{h^2}{\mu}, \tag{1.5}$$

where ν (the *true anomaly*) is the angle from $\boldsymbol{e}$ to $\boldsymbol{r}$. This is a conic section with the central body at one focus: $e = 0$ circle, $0 < e < 1$ ellipse, $e = 1$ parabola, $e > 1$ hyperbola. The eccentricity is not a new constant: squaring $\boldsymbol{e} = (\boldsymbol{v} \times \boldsymbol{h})/\mu - \hat{\boldsymbol{r}}$ and using $|\boldsymbol{v} \times \boldsymbol{h}| = vh$ (since $\boldsymbol{v} \perp \boldsymbol{h}$) and $\hat{\boldsymbol{r}} \cdot (\boldsymbol{v} \times \boldsymbol{h}) = h^2/r$ gives

$$e^2 = 1 + \frac{2\mathcal{E}h^2}{\mu^2}, \tag{1.6}$$

so the two conserved scalars fix the shape: $\mathcal{E} < 0$ gives $e < 1$, $\mathcal{E} = 0$ gives $e = 1$, $\mathcal{E} > 0$ gives $e > 1$, consistent with the classification of Section 1.2. (The lunar module uses Eq. (1.6) to get the perilune hyperbola's eccentricity from $\mathcal{E}_2$ and h_2.) The closest point ($\nu = 0$, *periapsis*) and, for an ellipse, the farthest ($\nu = \pi$, *apoapsis*) are

$$r_p = \frac{p}{1+e}, \qquad r_a = \frac{p}{1-e}, \qquad a = \frac{r_p + r_a}{2} = \frac{p}{1-e^2}, \qquad e = \frac{r_a - r_p}{r_a + r_p}. \tag{1.7}$$

Evaluating Eq. (1.4) at periapsis with $v_p = h/r_p$ gives the relation that ties energy to size,

$$\mathcal{E} = -\frac{\mu}{2a} \qquad \Longrightarrow \qquad v^2 = \mu\left(\frac{2}{r} - \frac{1}{a}\right) \quad \text{(vis-viva)}, \tag{1.8}$$

valid for every conic with the convention $a < 0$ for a hyperbola ($a \to \infty$ for a parabola). Figure 1.2 collects the geometry.

1.4 Speeds: circular, escape, hyperbolic

From Eq. (1.8):

$$v_c = \sqrt{\frac{\mu}{r}}\ (a = r), \qquad v_{esc} = \sqrt{\frac{2\mu}{r}} = \sqrt{2}\,v_c\ (\mathcal{E} = 0), \qquad v = \sqrt{v_\infty^2 + \frac{2\mu}{r}}\ (\mathcal{E} = \tfrac{1}{2}v_\infty^2 > 0). \tag{1.9}$$

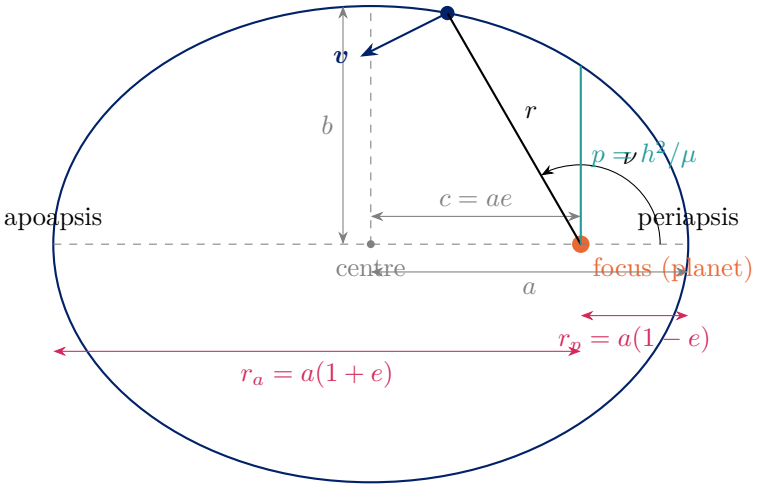


Figure 1.2: Geometry of an elliptical orbit ($e = 0.66$). The planet sits at a focus, not the centre; p is the radius at $\nu = 90°$.

The third form—speed on a hyperbola at radius r for a given excess speed—is the single most used formula in interplanetary Δv estimates. Figure 1.3 plots all three with an elliptical example.

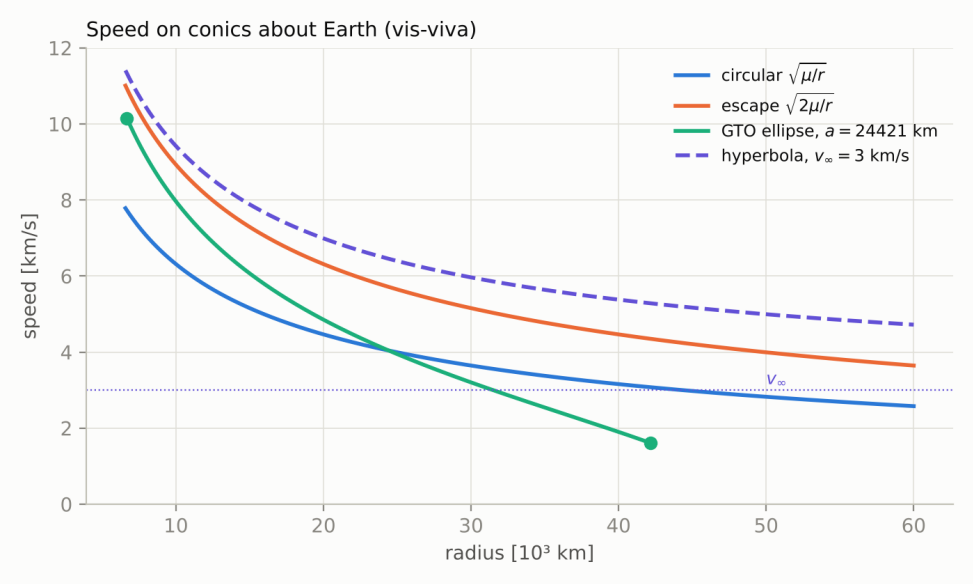


Figure 1.3: Speed versus radius about Earth for circular, escape, an elliptical (GTO) and a hyperbolic orbit. Where the GTO curve crosses the circular curve is where a circular orbit of that radius has the same speed—nowhere useful; the ellipse is faster than circular at perigee and slower at apogee.

Worked example – ISS and GTO

A 420-km circular orbit: $r = 6798\,\text{km}$, $v_c = 7.657\,\text{km/s}$, period $T = 2\pi\sqrt{r^3/\mu} = 93.0\,\text{min}$, escape speed $10.829\,\text{km/s}$, $\mathcal{E} = -29.32\,\text{km}^2/\text{s}^2$. A geostationary transfer orbit (GTO) with perigee 300 km and apogee 35,786 km: $a = 24\,421\,\text{km}$, $e = 0.7265$, $p = 11\,530\,\text{km}$, $h = 67\,793\,\text{km}^2/\text{s}$; perigee speed $10.151\,\text{km/s}$, apogee speed $1.608\,\text{km/s}$ (vis-viva at the two radii), period $10.55\,\text{h}$, $\mathcal{E} = -8.161\,\text{km}^2/\text{s}^2$. Geostationary radius from $T = 86\,164\,\text{s}$: $a = (\mu T^2/4\pi^2)^{1/3} = 42\,164\,\text{km}$ (altitude $35\,786\,\text{km}$), $v_c = 3.075\,\text{km/s}$.

1.5 Time: Kepler's equation

Equation (1.5) gives shape, not timing. For an ellipse the period follows from Kepler's second law, $\mathrm{d}A/\mathrm{d}t = h/2$, integrated over one revolution (area πab with $b = a\sqrt{1-e^2}$ and $h = \sqrt{\mu p} = \sqrt{\mu a(1-e^2)}$):

$$T = \frac{2\pi ab}{h} = 2\pi\sqrt{\frac{a^3}{\mu}}, \qquad n \equiv \frac{2\pi}{T} = \sqrt{\frac{\mu}{a^3}} \quad \text{(mean motion; Kepler's third law).} \tag{1.10}$$

To know *where* the spacecraft is at time t one introduces the *mean anomaly* $M = n(t - t_p)$ (an angle that grows uniformly with time, t_p the periapsis passage) and the *eccentric anomaly* E (an auxiliary angle measured from the ellipse's centre). They are linked by

$$M = E - e\sin E \quad \text{(Kepler's equation)}, \qquad \tan\frac{\nu}{2} = \sqrt{\frac{1+e}{1-e}}\,\tan\frac{E}{2}, \qquad r = a(1 - e\cos E). \tag{1.11}$$

Kepler's equation is transcendental: given M it is solved for E by Newton iteration ($E_{k+1} = E_k - (E_k - e\sin E_k - M)/(1 - e\cos E_k)$), which converges in a handful of steps. For a hyperbola the analogue is $M = e\sinh F - F$ with $r = a(1 - e\cosh F)$. Figure 1.4 shows the effect: on an eccentric orbit the true anomaly races through periapsis and crawls through apoapsis.

Worked example – Time on the GTO

From perigee to $\nu = 90°$ on the GTO above: $E = 2\arctan\left(\sqrt{(1-e)/(1+e)}\tan 45°\right) = 43.40°$, $M = E - e\sin E = 14.80°$, $t = M/n = 26.0\,\text{min}$—against $316.5\,\text{min}$ for half the orbit. The spacecraft spends most of its time near apogee.

Universal variables. Code that must handle ellipses, parabolas and hyperbolas with one formula uses the *universal anomaly* χ and the Stumpff functions $C(z)$, $S(z)$ of $z = \chi^2/a$; Kepler's equation becomes a single root-finding problem for all conics and the position at time t follows from the Lagrange coefficients f, g (Chapter 5). The derivation is in Curtis Ch. 3 and Bate–Mueller–White Ch. 4; for this tutorial it is enough to know that `propagate(r0, v0, dt, mu)` does exactly this.

1.6 Orbits in three dimensions

A conic in space needs six numbers (the *classical orbital elements*): size a, shape e, and three angles that orient the orbital plane and the periapsis—inclination i of the plane to the reference

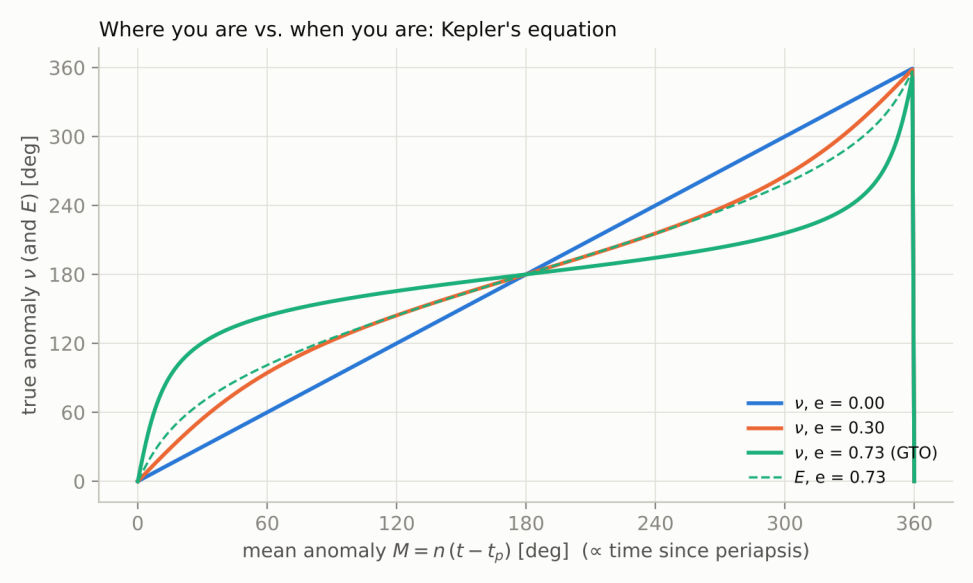


Figure 1.4: True anomaly (position) versus mean anomaly (time) for three eccentricities. On the GTO ($e = 0.73$) the spacecraft covers the first 90° of true anomaly in 26.0 min of a 10.55 h period.

plane, right ascension of the ascending node Ω (where the orbit crosses the reference plane going north), argument of periapsis ω (from the node to periapsis)—plus the true anomaly ν for the position. The reference plane is the Earth's equator (J2000) for Earth orbits and the ecliptic (Earth's orbital plane) for heliocentric work; the two are tilted by the obliquity 23.44°. Converting a state vector $(\boldsymbol{r}, \boldsymbol{v})$ to elements is a sequence of cross and dot products ($\boldsymbol{h} = \boldsymbol{r} \times \boldsymbol{v}$, node vector $\boldsymbol{N} = \hat{\boldsymbol{K}} \times \boldsymbol{h}$, eccentricity vector from the derivation above); converting back rotates the in-plane position $r(\cos\nu, \sin\nu)$ through ω, i, Ω. For mission analysis we spend most of our time with magnitudes in a plane, but the three-dimensional machinery is what lets Chapters 4 and 5 place real planets in real frames.

Where this lives in `mission_analysis`

`kepler.py`: `vis_viva`, `circular_velocity`, `escape_velocity`, `period`, `specific_energy`; `kepler_E`, `kepler_H`, `time_since_periapsis`; `coe_from_rv`, `rv_from_coe` (elements ↔ state); `propagate` (universal-variable two-body propagation); hyperbola helpers `hyperbolic_periapsis_speed`, `hyperbolic_eccentricity`, `turn_angle`.

Pitfall

The planet is at the *focus*, so the distance from the planet to the orbit's centre is $c = ae$, and periapsis radius is $a(1 - e)$, not a. Altitude is radius minus the body radius; forgetting to add R when converting an altitude is the most common error in this subject.

Exercises

1.1 Circular speed, period and escape speed for a 500-km orbit about Mars.

1.2 A burn at a 300-km Earth perigee raises speed by 3.5 km/s above circular. Is the orbit bound? If not, what is v_∞ and $C_3 = v_\infty^2$?

1.3 On the GTO of the worked example, how long from perigee to apogee? Why is it exactly half the period regardless of e?

Chapter 2

Changing orbits: impulsive maneuvers

2.1 The impulsive approximation

A chemical burn of a few minutes on an orbit of ninety minutes (or of a few hours on a transfer of months) can be idealized as an instantaneous change of velocity at fixed position: the *impulsive* Δv. Before the burn the spacecraft is on conic 1 at $\boldsymbol{r}$ with $\boldsymbol{v}_1$; after it, on conic 2 at the same $\boldsymbol{r}$ with $\boldsymbol{v}_2$; the propulsion system must deliver

$$\Delta v = |\boldsymbol{v}_2 - \boldsymbol{v}_1|. \tag{2.1}$$

The approximation fails when the burn is long compared with the time over which gravity changes the velocity direction—low thrust-to-weight upper stages lose a few percent (*gravity loss*), electric propulsion is a different regime entirely (not covered here).

For the propulsion engineer

Mission analysis ends where propulsion design starts: the rocket equation $\zeta = m_p/m_0 = 1 - e^{-\Delta v/c}$ with $c = g_0 I_{sp}$ (Chapter 9). A budget of 5.7 km/s is 0.837 of the initial mass in propellant at $I_{sp} = 320$ s and 0.725 at 450 s. Every 100 m/s saved in this tutorial is worth about 3% of the stage's initial mass—that is why the subject is worth a propulsion engineer's time.

2.2 Energy per unit Δv: the Oberth effect

Differentiating Eq. (1.4) at fixed r: $\mathrm{d}\mathcal{E} = \boldsymbol{v} \cdot \mathrm{d}\boldsymbol{v}$. A given Δv changes the orbital energy most when it is applied *along* the velocity and where the velocity is *largest*—at periapsis. The same propellant buys more energy deep in a gravity well. This is the Oberth effect and it decides where interplanetary injections and captures are performed (Section 3.2).

2.3 The Hohmann transfer

Between two coplanar circular orbits of radii $r_1 < r_2$ the two-impulse transfer of minimum total Δv is the ellipse tangent to both: periapsis r_1, apoapsis r_2, $a_t = (r_1 + r_2)/2$. With vis-viva on the three orbits,

$$v_{c,1} = \sqrt{\mu/r_1}, \qquad v_{t,p} = \sqrt{\mu\Big(\frac{2}{r_1} - \frac{1}{a_t}\Big)}, \qquad \Delta v_1 = v_{t,p} - v_{c,1}, \tag{2.2}$$

$$v_{c,2} = \sqrt{\mu/r_2}, \qquad v_{t,a} = \sqrt{\mu\Big(\frac{2}{r_2} - \frac{1}{a_t}\Big)}, \qquad \Delta v_2 = v_{c,2} - v_{t,a}, \tag{2.3}$$

$$t_H = \pi\sqrt{a_t^3/\mu} \quad \text{(half the transfer-orbit period)}. \tag{2.4}$$

Both burns are tangential (no direction change), which is why the transfer is cheap. Normalized by $v_{c,1}$ the total depends only on r_2/r_1 (Fig. 2.1): it rises to a maximum of $0.5363\,v_{c,1}$ at $r_2/r_1 = 15.57$ and then *falls* toward the escape value $\sqrt{2}-1 = 0.4142$—raising an orbit to a very high circular altitude costs more than leaving altogether. For $r_2/r_1 \gtrsim 12$ a three-impulse bi-elliptic transfer (out to a high intermediate apoapsis and back) beats Hohmann slightly, at the cost of a much longer flight.

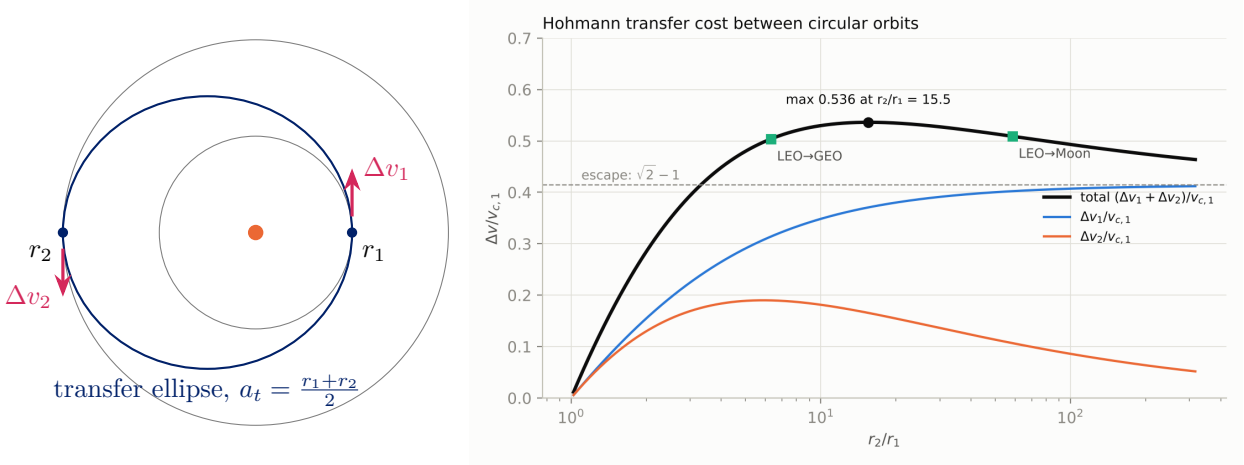


Figure 2.1: Left: Hohmann geometry; both burns are along the velocity. Right: normalized cost versus radius ratio, with the maximum at $r_2/r_1 \approx 15.57$ and the escape limit.

Worked example – LEO to GEO

$r_1 = 6678\,\text{km}$ (300 km), $r_2 = 42\,164\,\text{km}$: $v_{c,1} = 7.726\,\text{km/s}$, $v_{t,p} = 10.151\,\text{km/s}$, $\Delta v_1 = 2.426\,\text{km/s}$; $v_{t,a} = 1.608\,\text{km/s}$, $v_{c,2} = 3.075\,\text{km/s}$, $\Delta v_2 = 1.467\,\text{km/s}$; total $3.893\,\text{km/s}$, $t_H = 5.28\,\text{h}$. Bi-elliptic check at $r_2/r_1 = 15$ with an intermediate apoapsis of $60r_1$: $3.994\,\text{km/s}$ against Hohmann's $4.046\,\text{km/s}$, but 13.4 days instead of 18.3 h.

2.4 Plane changes

Rotating the velocity vector by Δi at constant speed costs

$$\Delta v = 2v \sin\frac{\Delta i}{2}, \tag{2.5}$$

which is proportional to v: do plane changes where the spacecraft is slowest. Combining a plane change with a speed change in one burn costs $\Delta v = \sqrt{v_1^2 + v_2^2 - 2v_1 v_2 \cos\Delta i}$, less than the two done separately. For LEO→GEO from a launch site at latitude 28.5° (which fixes the minimum inclination, Chapter 7) the 28.5° plane change costs $3.803\,\text{km/s}$ if done in LEO, $1.514\,\text{km/s}$ if done at GEO, and the optimal split—2.20° at the perigee burn, 26.30° at the apogee burn—brings the total transfer to $4.231\,\text{km/s}$ against $3.893\,\text{km/s}$ without plane change.

2.5 Phasing: when to go

A Hohmann transfer only arrives at the target if the target is there. The target must lead the departure point by the angle it will *not* cover during the flight,

$$\phi_{lead} = \pi - n_2\, t_H, \tag{2.6}$$

and this geometry repeats every *synodic period* S, $1/S = 1/T_1 - 1/T_2$. For Earth–Mars: $\phi_{lead} = 44.3°$ and $S = 779.9$ days (2.14 yr). This is why Mars launch windows come every 26 months and why "when" is as much a design variable as "how fast".

Where this lives in `mission_analysis`

`transfers.py`: `hohmann`, `bielliptic`, `plane_change_dv`, `combined_dv`, `hohmann_with_plane_change` (optimal split by 1-D minimization), `synodic_period`, `hohmann_phase_angle`, `wait_for_phase`, `hohmann_round_trip`.

Pitfall

Δv is a vector difference. Adding magnitudes is only right when the two velocities are parallel (tangential burns). A plane change of 60° costs a full v—as much as the orbital speed itself.

Exercises

2.1 Hohmann transfer from a 400-km Earth orbit to the Moon's distance (384 400 km), treating the Moon as absent: Δv_1, Δv_2 to circularize there, and t_H. Compare Δv_1 with the TLI of Chapter 6.

2.2 Why does $\Delta v_2 / v_{c,1}$ in Fig. 2.1 peak and then decrease with r_2/r_1?

2.3 Show that combining a plane change with the apogee burn of a GTO is cheaper than doing it at perigee, using the numbers of the worked example.

Chapter 3

Leaving one body for another: patched conics

3.1 Spheres of influence

With the Sun, a planet and a spacecraft we have a three-body problem, which has no closed-form solution. The engineering escape is to notice that near a planet the planet's gravity dominates, far from it the Sun's does, and to *patch* two-body conics at the boundary. Laplace's sphere of influence (SOI), the radius where the planet's perturbation of the heliocentric motion equals the Sun's perturbation of the planetocentric motion, is

$$r_{SOI} = a\left(\frac{m}{M}\right)^{2/5}, \tag{3.1}$$

with a the planet's orbital radius and m/M its mass ratio to the Sun. Values: Venus 616×10^3 km, Earth 929×10^3 km ($= 0.0062$ AU, $145 R_\oplus$), Mars 577×10^3 km, Jupiter $48\,210 \times 10^3$ km, Saturn $54\,551 \times 10^3$ km; the Moon about Earth 66×10^3 km. Two consequences: on the heliocentric scale Earth's SOI is a point (0.6 % of an AU), so the spacecraft "leaves Earth" at Earth's position with Earth's velocity plus whatever excess velocity it has; on the planetocentric scale the SOI is effectively infinity (145 Earth radii), so the planetocentric conic can be treated as reaching $r \to \infty$. The SOI radius itself never appears in a Δv formula—only the two limits do.

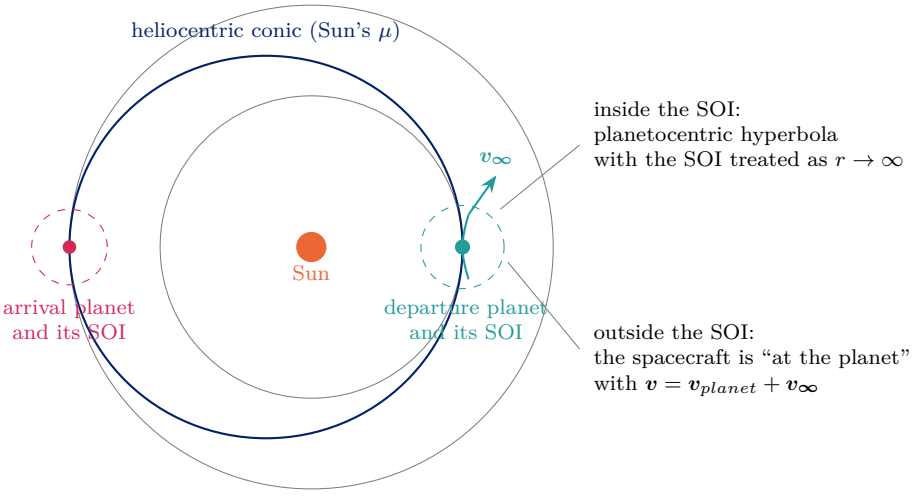


Figure 3.1: The patched-conic idea: a heliocentric conic between the planets' positions, and a planetocentric hyperbola inside each sphere of influence, matched through $\boldsymbol{v}_\infty$.

3.2 Departure: from parking orbit to hyperbola

Suppose the heliocentric leg requires the spacecraft to leave Earth's SOI with velocity $\boldsymbol{v}_\infty$ relative to Earth (how we find it is the subject of Section 3.4 and Chapter 5). Inside the SOI the spacecraft is on a hyperbola of energy $\mathcal{E} = v_\infty^2/2$. If it starts in a circular parking orbit of radius r_p and burns tangentially at the point that becomes the hyperbola's periapsis, Eq. (1.9) gives

$$\Delta v_{dep} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - \sqrt{\frac{\mu}{r_p}}, \qquad C_3 \equiv v_\infty^2. \tag{3.2}$$

C_3 (km^2/s^2) is the quantity launch providers quote: a launch vehicle's performance curve is payload mass versus C_3. The hyperbola's eccentricity, turning angle and asymptote direction (Fig. 3.2) are

$$e = 1 + \frac{r_p v_\infty^2}{\mu}, \qquad \delta = 2\arcsin\frac{1}{e}, \qquad \beta = \arccos\frac{1}{e}, \qquad \Delta = r_p\sqrt{1 + \frac{2\mu}{r_p v_\infty^2}}, \tag{3.3}$$

where β is the angle from the periapsis direction to the outgoing asymptote (it tells you where on the parking orbit to burn so that $\boldsymbol{v}_\infty$ points the right way) and Δ is the aiming radius—the lateral offset of the incoming asymptote from the planet's centre, used to target a flyby.

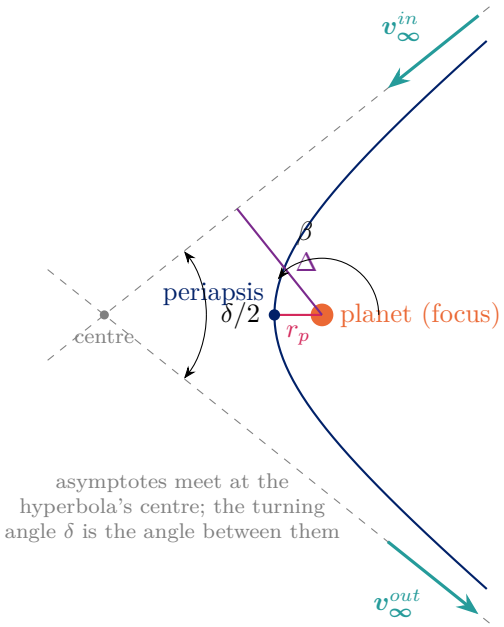


Figure 3.2: Hyperbolic passage ($e = 1.28$): incoming and outgoing asymptotes, turning angle $\delta = 2\arcsin(1/e)$, angle $\beta = \arccos(1/e)$ from periapsis to the outgoing asymptote, periapsis radius r_p and aiming radius Δ.

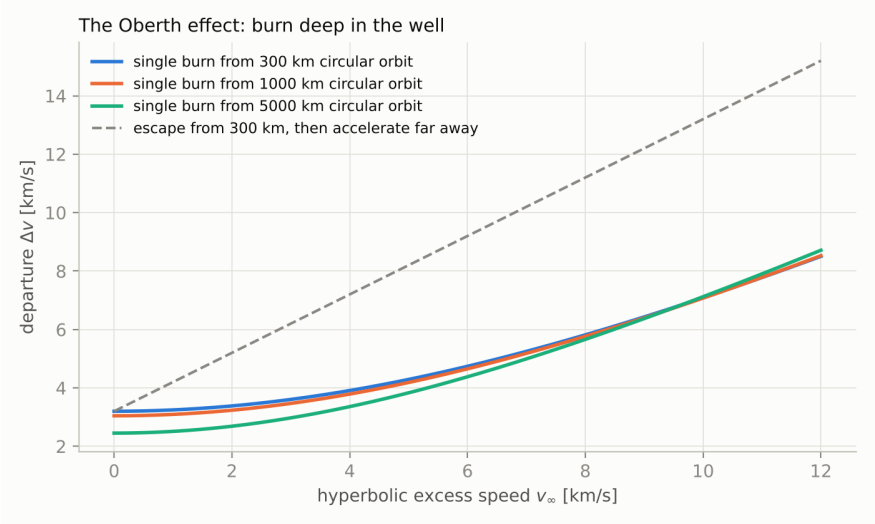


Figure 3.3: Departure Δv from a parking orbit versus v_∞ (Eq. (3.2)) compared with escaping first and accelerating later—the Oberth effect.

Worked example – Oberth in numbers

To leave Earth with $v_\infty = 3.0\,\mathrm{km/s}$ from a 300-km orbit ($v_c = 7.726\,\mathrm{km/s}$, $v_{esc} = 10.926\,\mathrm{km/s}$): a single tangential burn costs $\Delta v = \sqrt{3^2 + 2\mu/r_p} - v_c = 3.604\,\mathrm{km/s}$. Escaping first (3.200 km/s) and then accelerating to 3 km/s far from Earth would cost 6.2 km/s. The 3 km/s of excess speed is bought for 0.4 km/s because it is added where the spacecraft already moves at 11 km/s (Fig. 3.3).

3.3 Arrival: capture

Arriving with excess speed v_∞, the spacecraft is on a hyperbola. To stay it must burn at periapsis r_p from the hyperbolic speed down to the speed of the desired orbit:

$$\Delta v_{cap} = \sqrt{v_\infty^2 + \frac{2\mu}{r_p}} - \sqrt{\mu\Big(\frac{2}{r_p} - \frac{2}{r_p + r_a}\Big)}. \tag{3.4}$$

Capturing into a *high ellipse* ($r_a \gg r_p$) is much cheaper than circularizing, because the elliptical periapsis speed is close to the hyperbolic one; the remaining circularization can be left to repeated apoapsis-lowering burns or to aerobraking (Chapter 8). For circular capture there is an optimal periapsis: minimizing Eq. (3.4) with $r_a = r_p$ over r_p gives $r_p^* = 2\mu/v_\infty^2$ (Fig. 3.4)—at Mars with $v_\infty = 2.649\,\mathrm{km/s}$ that is an altitude of 8811 km, far above where science orbits are, so in practice one captures low into an ellipse.

3.4 The heliocentric leg: Hohmann between planets

Treat the planets' orbits as coplanar circles of radii a_1, a_2 about the Sun. The Hohmann ellipse between them has perihelion speed $v_{t,p}$ and aphelion speed $v_{t,a}$ (Chapter 2 with $\mu_\odot$). The planets move at $v_1 = \sqrt{\mu_\odot/a_1}$ and $v_2 = \sqrt{\mu_\odot/a_2}$ along the same directions, so the required excess speeds

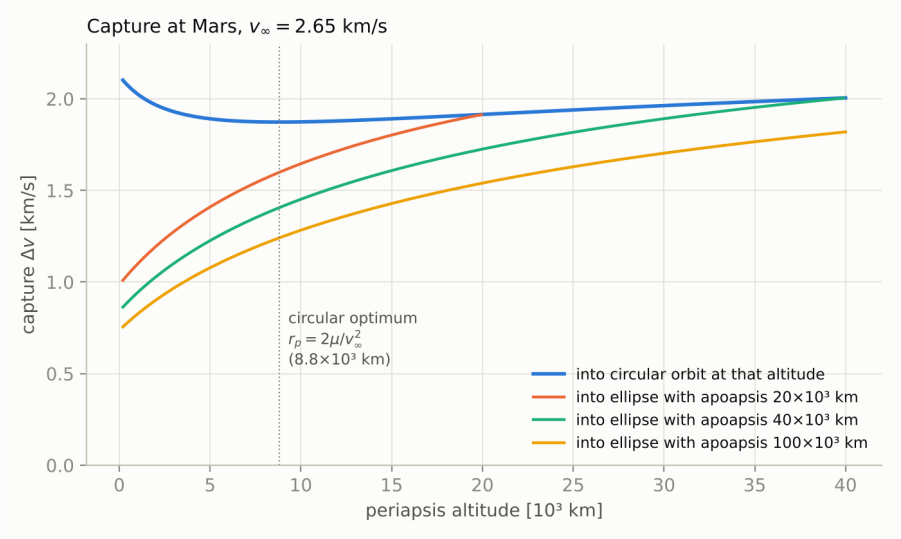


Figure 3.4: Capture Δv at Mars versus periapsis altitude for circular and elliptical capture orbits.

are simply

$$v_\infty^{dep} = v_{t,p} - v_1, \qquad v_\infty^{arr} = v_2 - v_{t,a}. \tag{3.5}$$

These feed Eqs. (3.2) and (3.4). Table 3.1 gives the result for four targets; these are the "textbook" numbers against which any dated mission (Chapter 5) should be checked.

Worked example – Earth to Mars, textbook

$v_{t,p} = 32.729\,\text{km/s}$ vs. Earth's $29.785\,\text{km/s} \Rightarrow v_\infty^{dep} = 2.945\,\text{km/s}$ ($C_3 = 8.67\,\text{km}^2/\text{s}^2$); $v_{t,a} = 21.480\,\text{km/s}$ vs. Mars's $24.129\,\text{km/s} \Rightarrow v_\infty^{arr} = 2.649\,\text{km/s}$; $t_H = 258.9$ days. Departure from a 300-km orbit: hyperbolic periapsis speed $11.316\,\text{km/s}$, $\Delta v = 3.590\,\text{km/s}$, $e = 1.145$, $\beta = 29.2°$. Capture at Mars into a 400-km circular orbit: hyperbolic speed $5.439\,\text{km/s}$, circular $3.359\,\text{km/s}$, $\Delta v = 2.080\,\text{km/s}$; into a $400 \times 40\,000$ km ellipse: $\Delta v = 0.884\,\text{km/s}$.

Table 3.1: Hohmann (circular, coplanar) transfers from Earth, departure from a 300-km parking orbit.

Target	v_∞^{dep} [km/s]	C_3 [km^2/s^2]	Δv_{dep} [km/s]	v_∞^{arr} [km/s]	t_H [d]	lead angle	synodic [yr]
Venus	2.50	6.2	3.48	2.71	146	−54.0°	1.60
Mars	2.94	8.7	3.59	2.65	259	44.3°	2.14
Jupiter	8.79	77.3	6.30	5.64	998	97.2°	1.09
Saturn	10.29	105.9	7.28	5.44	2208	106.1°	1.04

3.5 Gravity assists

Inside a planet's SOI the spacecraft's speed relative to the planet is the same on the way in and out ($|\boldsymbol{v}_\infty^{in}| = |\boldsymbol{v}_\infty^{out}|$: energy is conserved in the planet frame), but its direction is rotated by the turning angle δ. In the *Sun's* frame the velocity is $\boldsymbol{v} = \boldsymbol{v}_{planet} + \boldsymbol{v}_\infty$, and because $\boldsymbol{v}_\infty$ has rotated, $|\boldsymbol{v}|$ changes (Fig. 3.5). The free heliocentric speed change is

$$|\Delta v_{helio}| = |\boldsymbol{v}_\infty^{out} - \boldsymbol{v}_\infty^{in}| = 2v_\infty \sin\frac{\delta}{2} = \frac{2v_\infty}{1 + r_p v_\infty^2/\mu}. \tag{3.6}$$

For a fixed closest approach r_p this is largest when $v_\infty = \sqrt{\mu/r_p}$—the circular speed at the flyby radius—and the maximum equals that speed. Passing *behind* the planet (trailing side) adds energy; passing in front subtracts it. The momentum comes from the planet, whose orbit changes immeasurably. When the heliocentric legs before and after the flyby demand $\boldsymbol{v}_\infty$ vectors of different magnitude, the mismatch is made up by a small burn at periapsis (a *powered flyby*); `flyby_powered` finds the shared periapsis radius from the required turning and charges the speed difference.

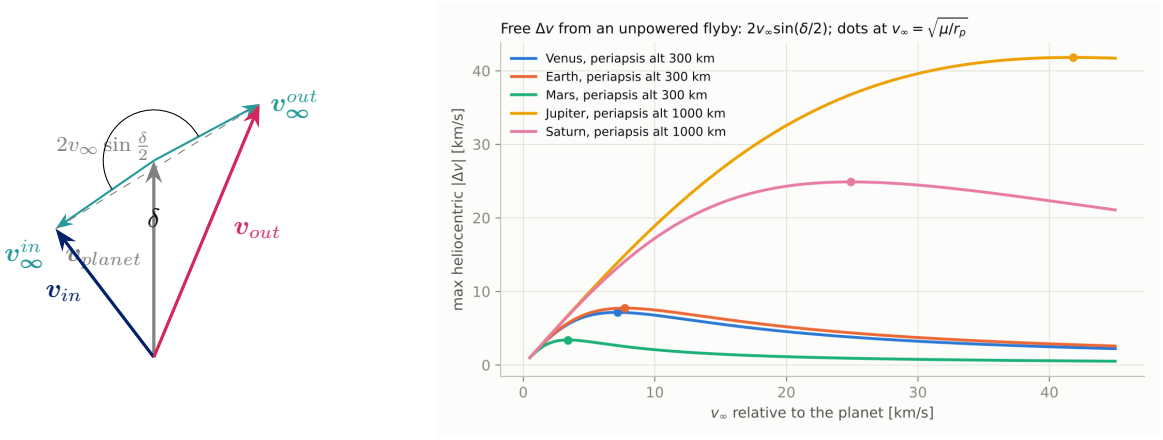


Figure 3.5: Left: in the planet frame $\boldsymbol{v}_\infty$ rotates by δ at constant magnitude; adding $\boldsymbol{v}_{planet}$ gives a longer heliocentric velocity. Right: maximum free Δv versus v_∞ for five planets at a fixed closest approach.

Worked example – Voyager 2 at Jupiter

Using only the launch (1977-08-20), Jupiter (1979-07-09) and Saturn (1981-08-25) dates and the ephemeris of Chapter 4, the Earth–Jupiter and Jupiter–Saturn legs (Chapter 5) give $v_\infty^{in} = 7.898\,\text{km/s}$ and $v_\infty^{out} = 7.775\,\text{km/s}$ at Jupiter, a required turning of 97.0°, a closest approach of $692 \times 10^3\,\text{km}$ (the real flyby was about $650 \times 10^3\,\text{km}$ from the centre), and a heliocentric speed change from 9.64 km/s to 19.65 km/s—11.74 km/s for a powered correction of only 0.047 km/s. Launch C_3 was $104.7\,\text{km}^2/\text{s}^2$.

Where this lives in `mission_analysis`

`patched_conics.py`: `departure_dv`, `capture_dv`, `optimal_capture_periapsis`, `flyby_unpowered`, `flyby_powered`, `flyby_max_heliocentric_dv`, `hohmann_interplanetary` (Table 3.1), `heliocentric_leg`, `interplanetary_mission` (a whole sequence of bodies and dates). SOI radii: `Body.soi`.

Pitfall

v_∞ is a velocity *relative to the planet*; C_3 is its square. A heliocentric speed is never a v_∞. And the departure Δv is *not* v_∞: it is Eq. (3.2), which for small v_∞ is close to the escape burn $(\sqrt{2}-1)v_c$.

Exercises

3.1 Jupiter arrival with the Hohmann v_∞ of Table 3.1: capture Δv into a 1000-km circular orbit, and into a $1000 \times 2 \times 10^6$ km ellipse. Where is the circular optimum?

3.2 A Venus flyby at 300-km altitude with $v_\infty = 5\,\mathrm{km/s}$: eccentricity, turning angle, maximum heliocentric Δv.

3.3 Explain with Fig. 3.5 why a flyby cannot change $|\boldsymbol{v_\infty}|$ but can change the spacecraft's heliocentric energy.

Chapter 4

Where are the planets? Ephemerides

Chapter 3 assumed circular coplanar planetary orbits. Real missions launch on calendar dates, so we need planet positions and velocities as functions of time: an *ephemeris*.

4.1 Mean elements and secular rates

A planet's orbit is nearly Keplerian; the other planets perturb its elements slowly. Over a few centuries the drift is well represented by linear trends, and JPL publishes exactly such a table (*Keplerian Elements for Approximate Positions of the Major Planets*): for each planet six elements at the epoch J2000 and six rates per Julian century, valid 1800–2050 (a second table extends to 3000 BC–3000 AD). The algorithm is the inverse of Chapter 1:

1. $T = (JD_{TT} - 2451545.0)/36525$ centuries since J2000;
2. $a, e, i, L, \varpi, \Omega$ each $= x_0 + \dot{x}\,T$ (the table uses mean longitude L and longitude of perihelion ϖ; then $\omega = \varpi - \Omega$ and $M = L - \varpi$);
3. solve Kepler's equation for E, then ν and r (Eq. (1.11));
4. rotate the in-plane position and velocity through ω, i, Ω into the J2000 ecliptic frame.

The result is good to tens of arcseconds for the inner planets and a few arcminutes for Saturn (JPL's own statement, confirmed in the package tests against an independent model). In velocity terms the error is of order 1 m/s at Earth—negligible for Δv budgets, unusable for navigation.

Worked example – Mars on the day of the lecture (2026-11-12)

$T = 0.268624$ centuries. Mars: $a = 1.523\,715$ AU, $e = 0.093415$, $M = 120.82°$, $\nu = 129.46°$; heliocentric distance 1.6058 AU, speed 22.863 km/s, ecliptic longitude 105.6°. Earth: 0.9900 AU, 30.085 km/s, longitude 49.2°. Mars leads Earth by 56.4°—more than the Hohmann lead angle of 44.3°, so the ideal departure date is *just past*; the porkchop plot of Chapter 5 confirms the window is open. Earth–Mars distance 1.341 AU.

4.2 Frames, the Moon and time

Planet positions are referred to the J2000 ecliptic; Earth-centric work (launch azimuths, the declination of a departure asymptote) uses the J2000 equator, a rotation by the obliquity 23.439° about the x-axis. The table's "Earth" is the Earth–Moon barycenter, which Earth itself orbits with an amplitude of 4700 km; the package can correct for this. The Moon's geocentric position comes from a truncated Meeus series (dozens of periodic terms in the Moon's mean anomaly, elongation and argument of latitude): on 2026-11-12 the Moon is 404 165 km from Earth moving at 0.975 km/s; its distance ranges 363,000–406,000 km over a month, which matters for lunar-transfer Δv at the 10 m/s level. Epochs are Julian Dates; the ephemeris wants Terrestrial Time, which runs 69 s ahead of UTC—an offset far below the accuracy of mean elements.

Where this lives in `mission_analysis`

`ephemeris.py`: `planet_elements`, `planet_state`, `earth_state` (barycenter correction), `moon_state_geocentric`, `body_state` (any body, one call), `heliocentric_longitude`, `phase_angle`, `to_jd`, `jd_to_iso`.

Pitfall

Mean elements are not osculating elements: the planet's instantaneous velocity is computed from the two-body relation on the *mean* orbit, which is what makes the model consistent with the Lambert solver that follows. Mixing in a precise ephemeris for one body and mean elements for another buys nothing.

Chapter 5

Lambert's problem and porkchop plots

5.1 The problem

Given two position vectors $\boldsymbol{r}_1$, $\boldsymbol{r}_2$ about a central body and a time of flight Δt, find the conic that connects them—i.e. the velocities $\boldsymbol{v}_1$ at departure and $\boldsymbol{v}_2$ at arrival. This is Lambert's problem, and it is the workhorse of mission analysis because the calendar, not the Hohmann geometry, fixes where the planets are: a departure on a given date and an arrival on another date defines $\boldsymbol{r}_1$ (Earth then), $\boldsymbol{r}_2$ (Mars then) and Δt. The Hohmann transfer is the special case where the two positions happen to be 180° apart and $\Delta t = t_H$.

5.2 Geometry

The transfer angle $\Delta\theta$ between $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ is either $< 180^\circ$ (*type I*, the "short way") or $> 180^\circ$ (*type II*, the "long way"); for a spacecraft that must move in the same sense as the planets (prograde), the sign of $(\boldsymbol{r}_1 \times \boldsymbol{r}_2)_z$ decides which. Exactly at $\Delta\theta = 180^\circ$ the two positions and the focus are collinear and the transfer plane is undefined—this singularity shows up as a ridge in every porkchop plot. For a given geometry the ellipse of minimum energy has $a_{min} = (r_1 + r_2 + c)/4$, c the chord; longer flights use larger ellipses, shorter ones hyperbolas.

5.3 How it is solved

Position and velocity at two times on the same conic are related by the Lagrange coefficients,

$$\boldsymbol{r}_2 = f\,\boldsymbol{r}_1 + g\,\boldsymbol{v}_1, \qquad \boldsymbol{v}_2 = \dot{f}\,\boldsymbol{r}_1 + \dot{g}\,\boldsymbol{v}_1, \tag{5.1}$$

so once f and g are known, $\boldsymbol{v}_1 = (\boldsymbol{r}_2 - f\boldsymbol{r}_1)/g$ and $\boldsymbol{v}_2 = (\dot{g}\boldsymbol{r}_2 - \boldsymbol{r}_1)/g$. In the universal-variable formulation f, g, $\dot{g}$ and the time of flight all depend on the geometry and on one unknown $z = \chi^2/a$ (negative for hyperbolas, zero for the parabola, positive for ellipses) through the Stumpff functions $C(z)$, $S(z)$:

$$A = \sin\Delta\theta\sqrt{\frac{r_1 r_2}{1-\cos\Delta\theta}}, \quad y(z) = r_1 + r_2 + A\,\frac{zS(z)-1}{\sqrt{C(z)}}, \quad \sqrt{\mu}\,\Delta t = \Big[\frac{y}{C}\Big]^{3/2} S + A\sqrt{y}, \tag{5.2}$$

$$f = 1 - \frac{y}{r_1}, \qquad g = A\sqrt{\frac{y}{\mu}}, \qquad \dot{g} = 1 - \frac{y}{r_2}. \tag{5.3}$$

The time of flight is a monotonic function of z on the single-revolution branch (Fig. 5.1), so solving Lambert's problem is a one-dimensional root find: bracket z between a hyperbolic value and $z \to 4\pi^2$ (the one-full-revolution limit) and let Brent's method converge. This formulation never diverges, costs 55 µs per solve and handles all conics with one code path—the reasons it was chosen for the package.

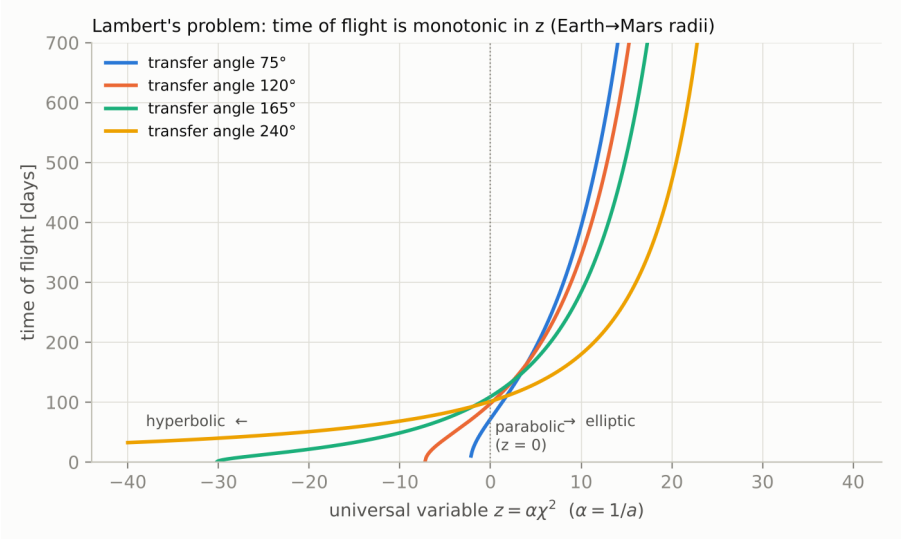


Figure 5.1: Time of flight versus the universal variable z for Earth–Mars radii and four transfer angles. Given Δt, the solver finds the z (hence the conic) by a bracketed root find.

5.4 From a Lambert solution to mission quantities

With $\boldsymbol{v}_1$ and $\boldsymbol{v}_2$ in hand and the planets' velocities from the ephemeris,

$$\boldsymbol{v}_\infty^{dep} = \boldsymbol{v}_1 - \boldsymbol{v}_{planet,1}, \qquad \boldsymbol{v}_\infty^{arr} = \boldsymbol{v}_2 - \boldsymbol{v}_{planet,2}, \tag{5.4}$$

and Eqs. (3.2) and (3.4) give the burns. The z-component of $\boldsymbol{v}_\infty^{dep}$ in the equatorial frame is the declination of the launch asymptote (DLA), which must be reachable from the launch site (Chapter 7).

Worked example – A dated Earth–Mars transfer

Depart 2026-10-30, arrive 2027-09-06 ($\Delta t = 311$ days). Earth at 0.9932 AU moving at 29.988 km/s; Mars at 1.5197 AU; transfer angle 206.2° (type II); transfer ellipse $a = 1.2693$ AU; $|\boldsymbol{v}_1| = 32.977$ km/s. Then $v_\infty^{dep} = 3.035$ km/s ($C_3 = 9.21$ km^2/s^2), $v_\infty^{arr} = 2.574$ km/s, departure Δv from 300 km 3.614 km/s, capture into 400 km circular 2.044 km/s. Compare the textbook Hohmann values of Chapter 3: the real geometry costs a few percent more.

5.5 Porkchop plots

Repeat the Lambert solution over a grid of departure dates and times of flight and contour the result: a *porkchop plot* (Fig. 5.2). Reading one is a core skill:

- the two lobes are type I (below the ridge, shorter flights) and type II (above it);
- the ridge between them is the 180° singularity;
- C_3 contours (what the launch vehicle must provide) and arrival v_∞ contours (what the capture system must absorb) have different optima—a mission chooses a compromise;
- the width of the "bowl" at an acceptable Δv is the *launch window*; it recurs every synodic period with a different shape, because the planets' eccentric, inclined orbits make some oppositions better than others.

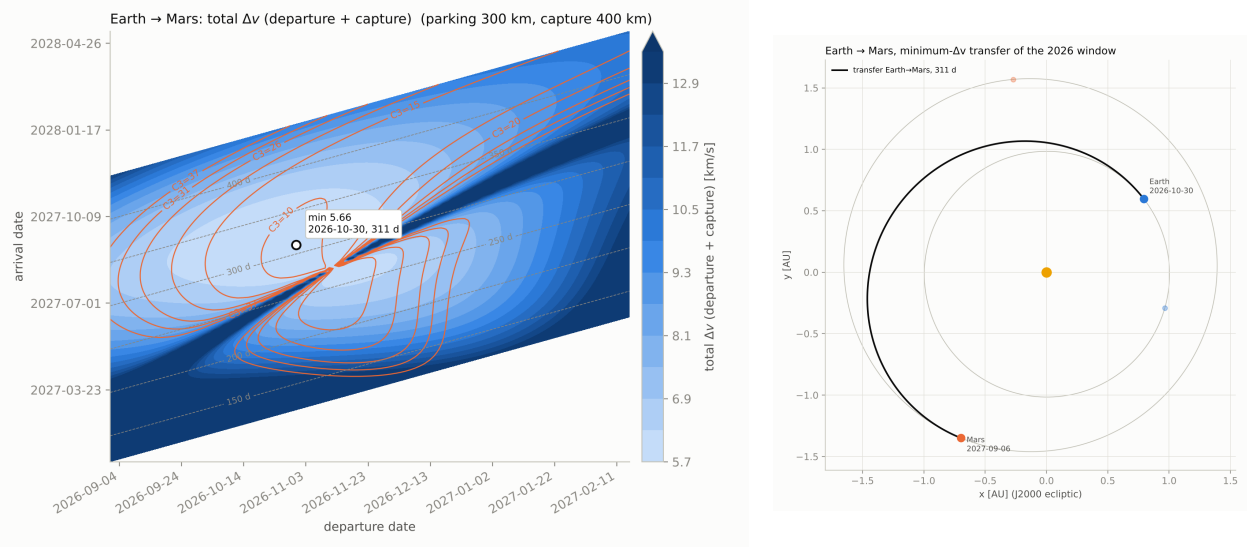


Figure 5.2: Earth→Mars 2026/27. Left: total Δv (300-km parking orbit to 400-km Mars orbit) with C_3 contours and time-of-flight lines. Right: the minimum-Δv transfer in plan view (type II, 206° of heliocentric travel).

Worked example – The window opening now

Grid: departures 2026-09-01 to 2027-02-15, flights 120–450 days. Minimum total Δv: depart 2026-10-30, 311 days, $C_3 = 9.22\,\mathrm{km^2/s^2}$, $v_\infty^{arr} = 2.572\,\mathrm{km/s}$, total 5.657 km/s (type II). Minimum C_3: $9.14\,\mathrm{km^2/s^2}$ on 2026-10-30 with a 293 day flight. Best type-I transfer: depart 2026-11-16, 260 days, $C_3 = 12.00$, $v_\infty^{arr} = 3.037\,\mathrm{km/s}$, total 6.015 km/s—0.36 km/s more for 51 days less. Window within 10% of the optimum: 2026-09-20 to 2026-11-30 (70 days). Check against reality: for Mars 2020 (launch 2020-07-30, entry 2021-02-18) the same machinery gives $C_3 = 14.50\,\mathrm{km^2/s^2}$, a 143.1° type-I transfer and a heliocentric path of 470.9 Mkm against the 471 Mkm in NASA's press kit.

Where this lives in `mission_analysis`

`lambert.py`: `lambert(r1, r2, tof, mu, prograde)` returning $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, the transfer angle and a. `porkchop.py`: `porkchop` (grids of C_3, v_∞^{arr}, Δv), `Porkchop.optimum`, `Porkchop.window`, `next_launch_windows` (multi-year scan). `plotting.plot_porkchop`, `plot_transfer`.

Pitfall

A porkchop plot answers "what does this date pair cost", not "is this trajectory flyable": it ignores the declination constraint, the arrival geometry (lighting, landing site) and navigation margins. It is the first filter, not the last.

Exercises

5.1 Solve the type-I leg 2026-11-16 → 2027-08-03 and report C_3, v_∞^{arr}, transfer angle.

5.2 Why is time of flight monotonic in z (Fig. 5.1) but not in a? (Hint: the minimum-energy ellipse.)

5.3 Explain why the porkchop lobes are bounded on the left by an almost vertical wall for short flights.

Chapter 6

The Moon: a patched conic at small scale

6.1 Why the Moon is different

The Moon's sphere of influence (66 183 km) is 17 % of the Earth–Moon distance $D = 384\,400$ km, and the Moon moves at 1.0232 km/s—comparable to a spacecraft's speed when it arrives. Neither "SOI is a point" nor "target is stationary" holds, so the patched conic must keep the geometry explicitly. It still works remarkably well.

6.2 Level 1: Hohmann to the Moon

Treat the Moon as a point at the apogee of a transfer ellipse from the parking orbit ($r_0 = 6563$ km for 185 km): $a_t = 195\,482$ km, injection speed $v_0 = 10.9283$ km/s against circular 7.793 km/s, so the trans-lunar injection (TLI) burn is $\Delta v_{TLI} = 3.135$ km/s with a flight of 4.98 days. At apogee the spacecraft moves at 0.1866 km/s, the Moon at 1.0232 km/s: the relative speed $v_\infty = 0.8366$ km/s becomes the excess speed of a selenocentric hyperbola, and Eq. (3.4) gives lunar orbit insertion into a 100-km circular orbit: $\Delta v_{LOI} = 2.457 - 1.634 = 0.823$ km/s. Total 3.959 km/s. These are the right magnitudes (Apollo flew 3.05–3.2 km/s TLI and 0.9 km/s LOI).

6.3 Level 2: the Bate–Mueller–White patched conic

Figure 6.1 defines the geometry. Inputs: the injection state—radius r_0, speed v_0, flight-path angle ϕ_0 (angle of the velocity above the local horizontal)—and the arrival angle λ_1, which places the point where the trajectory crosses the Moon's SOI.

1. **Geocentric conservation.** $\mathcal{E} = v_0^2/2 - \mu/r_0$ and $h = r_0 v_0 \cos\phi_0$ are constant on the geocentric leg.
2. **Arrival geometry.** The SOI point is at distance $r_1 = \sqrt{D^2 + r_{SOI}^2 - 2Dr_{SOI}\cos\lambda_1}$ from Earth, at an angle γ_1 from the Earth–Moon line with $\sin\gamma_1 = (r_{SOI}/r_1)\sin\lambda_1$.
3. **Arrival state.** $v_1 = \sqrt{2(\mathcal{E} + \mu/r_1)}$ and $\cos\phi_1 = h/(r_1 v_1)$ from the two conserved quantities.
4. **Patch.** The selenocentric velocity is the vector difference $\boldsymbol{v}_2 = \boldsymbol{v}_1 - \boldsymbol{v}_{Moon}$, with $\boldsymbol{v}_{Moon}$ perpendicular to the Earth–Moon line. Its magnitude and direction relative to the Moon-centred radius vector give $\mathcal{E}_2 = v_2^2/2 - \mu_M/r_{SOI}$ and $h_2 = |\boldsymbol{\rho}\times\boldsymbol{v}_2|$, hence $e_2 = \sqrt{1 + 2\mathcal{E}_2 h_2^2/\mu_M^2}$ (Eq. (1.6)) and the perilune radius $r_p = h_2^2/[\mu_M(1+e_2)]$. The sign of $(\boldsymbol{\rho}\times\boldsymbol{v}_2)_z$ says whether the spacecraft passes the Moon prograde or retrograde.
5. **Time.** Kepler's equation on the geocentric ellipse from r_0 to r_1, plus the hyperbolic Kepler equation from r_{SOI} to perilune (14–18 h—not negligible). The Moon's lead angle at TLI follows from the two times and ω_{Moon}.

In practice one wants a given perilune altitude; the package scans λ_1 (for a chosen v_0) or v_0 (for a chosen λ_1) and returns the solutions—usually one prograde and one retrograde.

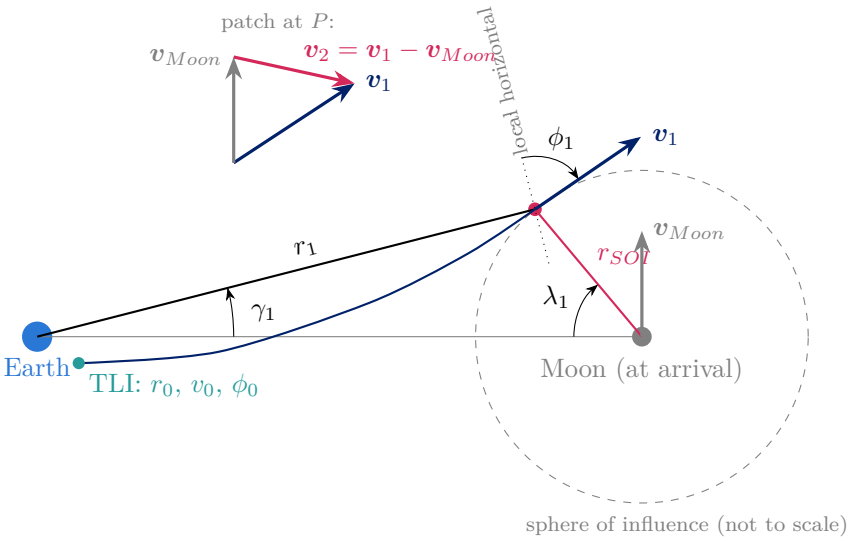


Figure 6.1: Bate–Mueller–White lunar patched conic. The arrival angle λ_1 fixes where the geocentric ellipse meets the Moon's sphere of influence; r_1 and γ_1 follow from the triangle Earth–Moon–P, the arrival speed and flight-path angle ϕ_1 from energy and angular momentum, and subtracting the Moon's velocity (inset) gives the selenocentric hyperbola.

Worked example – Two transfers with $v_0 = 10.95$ km/s from 185 km

Geocentric ellipse: $a = 255 \times 10^3$ km, $e = 0.9742$; TLI $\Delta v = 3.157$ km/s. *Retrograde solution* (Apollo-like): $\lambda_1 = 44.1°$, $r_1 = 340 \times 10^3$ km, $v_1 = 0.884$ km/s, $\phi_1 = 76.2°$, $\gamma_1 = 7.78°$; $v_\infty = 1.078$ km/s, $e_2 = 1.380$, perilune 100 km, LOI 0.886 km/s; 56.9 h to the SOI + 15.4 h to perilune = 72.3 h; Moon lead angle at TLI 131.8°. *Prograde solution*: $\lambda_1 = 35.9°$, $v_\infty = 1.108$ km/s, LOI 0.899 km/s, 69.8 h. Apollo 11's published TLI state (10.832 km/s at 336 km, flight-path angle $\approx 7°$) gives 67.8 h to perilune and LOI 0.919 km/s; the S-IVB cutoff row of the same table (10.840 km/s at 323 km) gives 69.5 h. Apollo 11 took 73.0 h and burned 0.891 km/s. Lunar time of flight moves about 5 h per 0.1 km^2/s^2 of injection energy, so the spread reflects the published altitude datum, not the method.

Where this lives in `mission_analysis`

`lunar.py`: `lunar_hohmann_estimate`; `lunar_patched_conic(r0, v0, phi0, lambda1)`; `design_lunar_transfer(park_alt, perilune_alt, v0=.. | lambda1_deg=..)`; `LunarPatchedConic.loi_dv`, `.tof_total`, `.direction`. `plotting.plot_lunar_transfer`.

Pitfall

Omitting the time from the SOI to perilune under-predicts lunar flight time by 15–20%; it is the first thing to check when a patched-conic lunar result disagrees with a flown mission.

Exercises

6.1 Find the retrograde solution for $v_0 = 11.0$ km/s from 185 km to a 100-km perilune: λ_1, total time, LOI.

6.2 Why does a faster transfer (larger v_0) cost more LOI even though TLI rises only slightly?

Chapter 7

Getting to orbit

Everything so far started in a parking orbit. The launch itself is a trajectory problem (gravity turn, drag, steering) treated in the Week-10 lecture and by the course's ascent-trajectory tool; mission analysis only needs a defensible Δv:

$$\Delta v_{launch} = \left|\boldsymbol{v}_{orbit} - \boldsymbol{v}_{site}\right| + \Delta v_{grav} + \Delta v_{drag} + \Delta v_{steer}. \tag{7.1}$$

The launch site already moves eastward with Earth's rotation, $v_{site} = \omega_{\oplus} R \cos\phi_L$ ($\omega_{\oplus} = 7.2921 \times 10^{-5}$ rad/s, 0.465 km/s at the equator). To reach inclination i from latitude ϕ_L the launch azimuth Az (from north) satisfies $\cos i = \cos\phi_L \sin Az$, which requires $i \geq \phi_L$: a site cannot reach an inclination lower than its latitude without a plane change. The ideal Δv is the vector difference between the orbital velocity at that azimuth and the eastward site velocity; losses are added as a band (gravity 1.0–1.5, drag 0.05–0.15, steering 0.1–0.4 km/s are typical for medium launchers).

Worked example – Kennedy Space Center ($\phi_L = 28.5°$) to 300 km

Due east ($i = 28.5°$): $v_{orbit} = 7.726$ km/s, $v_{site} = 0.409$ km/s, ideal $\Delta v = 7.317$ km/s, with losses 8.47–9.37 km/s. To the ISS inclination (51.6°, $Az = 45.0°$): credit 0.283 km/s, ideal 7.442 km/s. Polar: credit −0.011 km/s (slightly negative—the site velocity is now sideways), ideal 7.737 km/s. Kourou (5.2°) due east gains 0.463 km/s.

Where this lives in `mission_analysis`

`launch.py`: `launch_to_orbit(altitude, inclination_deg, latitude_deg, body, losses)`, `LossBand`, `dv_to_escape_from_orbit`.

Pitfall

The sign: $\Delta v = v_{orbit} + \text{losses} - v_{site}$. Earth's rotation *helps* an eastward launch. (The 2025 Week-9 slide had the signs reversed; the corrected estimate is 8.9–9.4 km/s with the slide's 1.5–2.0 km/s loss band.)

Chapter 8

Aero-assist: trading heat shield for propellant

8.1 Aerocapture

If the target has an atmosphere, one deep pass can remove the hyperbolic excess energy instead of the capture burn. The spacecraft enters at $v_{entry} = \sqrt{v_\infty^2 + 2\mu/r_{atm}}$ (5.598 km/s at Mars for $v_\infty = 2.649$ km/s), emerges on an ellipse whose periapsis is still in the atmosphere, and raises it with a small apoapsis burn (0.0637 km/s here). The budget credit is the whole propulsive capture, 2.080 km/s for a 400-km orbit—paid for with a heat shield, a guided entry and a narrow corridor of acceptable entry angles. The tool charges only the Δv side of that trade.

8.2 Aerobraking

Less demanding is to capture propulsively into a high ellipse (0.853 km/s into $110 \times 40\,000$ km at Mars) and lower the apoapsis over hundreds of shallow passes through the upper atmosphere. Each pass removes a little speed at periapsis. With an exponential atmosphere $\rho = \rho_p \exp[-(r - r_p)/H]$, drag deceleration $\mathrm{d}v/\mathrm{d}t = -\rho v^2/(2\beta)$ ($\beta = m/C_D A$ the ballistic coefficient) and the radius near periapsis expanded as $r - r_p \approx \frac{1}{2}\ddot{r}_p t^2$, the integral over a pass is Gaussian:

$$\Delta v_{pass} = \frac{\rho_p V_p^2}{2\beta}\sqrt{\frac{2\pi H}{\ddot{r}_p}}, \qquad \ddot{r}_p = \frac{V_p^2}{r_p} - \frac{\mu}{r_p^2}. \tag{8.1}$$

$\ddot{r}_p$ is the radial acceleration at periapsis: the centripetal term of the curved path minus gravity. Dropping gravity (a straight-line fly-through) gives the compact form $\rho_p V_p \sqrt{2\pi H r_p}/(2\beta)$ found on the lecture slide, which is low by the factor $\sqrt{V_p^2/(V_p^2 - \mu/r_p)}$—$\sqrt{2}$ for a near-parabolic pass, more for lower apoapses. Direct integration of the equations of motion with drag confirms Eq. (8.1) to 0.2% (Fig. 8.1). Pass after pass the apoapsis falls, periods shorten, and a final periapsis-raise burn circularizes.

Worked example – Mars aerobraking campaign

$\rho_p = 5.79 \times 10^{-8}$ kg/m^3 at 110 km, $H = 10.5$ km, $V_p = 4.754$ km/s, $\ddot{r}_p = 2.9630$ m/s^2, $\beta = 50$ kg/m^2: first pass $\Delta v = 1.95$ m/s (integration 1.951, gravity-aware 1.952, straight-line 1.323 m/s). From $110 \times 40\,000$ km to a circular 400-km orbit: 521 passes over 152 days, 1.192 km/s removed by drag, final raise 0.066 km/s. Propulsive total $0.853 + 0.066$ versus direct capture 2.080 km/s: saving 1.161 km/s. Halving the drag area (doubling β) doubles the passes and the campaign time.

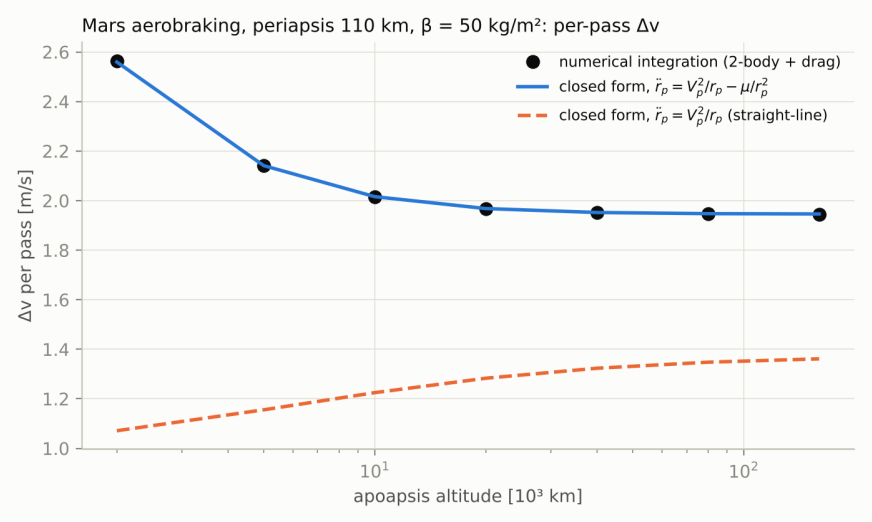


Figure 8.1: Per-pass Δv at Mars (periapsis 110 km, $\beta = 50\,\mathrm{kg/m^2}$): integration versus the two closed forms.

Where this lives in `mission_analysis`

`aerobraking.py`: `aerocapture_credit`, `aerobraking_pass_dv` (both closed forms), `aerobraking_pass_dv_numeric` (integrated check), `aerobraking_campaign`. Atmospheres: `Body.atmosphere` (exponential fits for the 90–150 km band).

Chapter 9

From Δv to mass: the rocket equation and staging

This is where the propulsion engineer takes over, so only the bridge is built here.

9.1 The rocket equation in mission terms

With $c = g_0 I_{sp}$ ($g_0 = 9.806\,65\,\mathrm{m/s^2}$) the propellant fraction for a burn or a whole budget is

$$\zeta = \frac{m_p}{m_0} = 1 - e^{-\Delta v/c}, \qquad \text{and inversely} \qquad {I_{sp}}^{req} = \frac{\Delta v}{g_0 \ln[1/(1-\zeta)]}. \tag{9.1}$$

For a 5.7 km/s budget: $\zeta = 0.837$ at 320 s (storable bipropellant), 0.725 at 450 s (hydrogen–oxygen), 0.476 at 900 s (nuclear thermal), 0.176 at 3000 s (electric—if the mission can afford the time). A single stage to orbit (9.3 km/s) with 90% propellant fraction needs $I_{sp} = 412\,\mathrm{s}$.

9.2 Staging

For stage i (burning first for $i = 1$) define the structural coefficient $\sigma_i = m_{s,i}/(m_{s,i} + m_{p,i})$, the stage mass ratio $R_i = m_{0,i}/(m_{0,i} - m_{p,i}) = e^{\Delta v_i/c_i}$ and the payload ratio $\pi_i = m_{0,i+1}/m_{0,i}$ (everything above stage i is its payload). Eliminating the stage mass,

$$\pi_i = \frac{1 - R_i\sigma_i}{R_i(1-\sigma_i)}, \qquad \frac{m_0}{m_L} = \prod_i \frac{1}{\pi_i} = \prod_i \frac{R_i(1-\sigma_i)}{1 - R_i\sigma_i}, \tag{9.2}$$

which requires $R_i\sigma_i < 1$: a stage with $\sigma_i = 0.08$ cannot deliver more than $c_i \ln(1/\sigma_i)$ even with zero payload. For a fixed total $\Delta v = \sum \Delta v_i$ the split that minimizes m_0 follows from a Lagrange multiplier λ: $R_i = (\lambda c_i - 1)/(\lambda c_i \sigma_i)$ with $\sum c_i \ln R_i = \Delta v$—one scalar root; identical stages give equal Δv_i. (The package names σ_i `structural_fraction` and reports $\lambda_i = m_{0,i+1}/(m_{s,i} + m_{p,i}) = \pi_i/(1-\pi_i)$ as `payload_ratio`.)

9.3 Budgets

A `DeltaVBudget` is the hand-off object: an ordered list of legs (launch with its loss band, injection, flybys, capture or aerocapture, landing, margin), each optionally tagged with the I_{sp} of the stage that performs it. From it: the post-launch propulsive Δv, the propellant fraction at a given I_{sp}, the required I_{sp} at a given fraction, and the propellant per leg by working backward from the final mass.

Worked example – HW 1 chain: a Mars orbiter in the 2026 window

Dates from the porkchop optimum (2026-10-30 → 2027-09-06). Legs: launch 8.917 km/s (band 8.47–9.37), injection 3.614 km/s, capture 2.043 km/s, 5% margin 0.729 km/s: total 15.303 km/s (13.225 km/s with aerocapture). Post-launch propulsive Δv: 6.386 km/s vs. 4.308 km/s. Required I_{sp} at $\zeta = 0.5/0.6/0.7$: 939/711/541 s propulsive, 634/479/365 s with aerocapture. A 1000 kg final mass with a 320-s engine (propulsive capture) needs 261 (margin) + 1157 (capture) + 5233 kg (injection) of propellant: 7652 kg injected onto the escape hyperbola. A two-stage launcher (I_{sp} 300/450 s, σ 0.07/0.10) delivering that mass through 12.531 km/s (launch + injection): optimal split 4.822/7.709 km/s, stage masses 601/85.0 t, gross 694 t (payload fraction 1.10 %); an equal split would weigh 924 t.

Where this lives in `mission_analysis`

`staging.py`: `tsiolkovsky_dv`, `propellant_fraction_for_dv`, `required_isp`, `design_vehicle` (optimal split), `size_vehicle`, `size_burn_sequence`. `budget.py`: `DeltaVBudget` (`table`, `size_propellant`, `required_isp`), `interplanetary_budget`, `lunar_budget`.

Chapter 10

Putting it together

The chain below is what HW 1 asks for and what the project model begins with; each line is one call into the package.

```
from modern_aerospace_propulsion.mission_analysis import *

# 1. textbook orientation (Ch. 2-3): circular coplanar Hohmann
print(hohmann_interplanetary("Earth", "Mars", park_alt=300, capture_peri_alt=400).summary())

# 2. the real window (Ch. 4-5): porkchop over dated Lambert solutions
pc = porkchop("Earth", "Mars", "2026-09-01", "2027-02-15", 120, 450,
              park_alt=300, capture_peri_alt=400)
opt = pc.optimum("dv_total")               # depart, arrive, C3, v_inf, dv

# 3. the budget (Ch. 7-9): launch band + injection + capture (or aerocapture) + margin
budget, mission = interplanetary_budget("Earth", "Mars", opt["depart"], opt["arrive"],
                                        park_alt=300, launch_lat_deg=28.5,
                                        capture_peri_alt=400, aerocapture=False)
print(budget.table())

# 4. propellant and staging (Ch. 9)
rows = budget.size_propellant(final_mass=1000.0, default_isp=320.0)
m_injected = rows[0]["m_before"]
launcher = design_vehicle(budget.legs[0].dv + budget.legs[1].dv, m_injected,
                          isps=[300, 450], sigmas=[0.07, 0.10])
print(launcher.table())

# 5. levers (Ch. 3, 8): aerocapture, elliptical capture, gravity assist
print(aerocapture_credit("Mars", mission.capture.vinf, 400).summary())
print(capture_dv("Mars", mission.capture.vinf, 400, 40000).summary())
```

Read the output against this tutorial: the Hohmann numbers of Table 3.1 bound the dated ones; the porkchop optimum sits on the type-II lobe; the capture burn is Eq. (3.4); the budget's propellant fraction is the rocket equation; and the launcher's optimal split is the Lagrange result of Chapter 9. When a number surprises you, the chapter that owns it is where to look.

What this model does not know. Patched conics ignore third-body gravity during each leg; burns are impulsive; the ephemeris is good to arcminutes; the Moon's orbit is taken circular; aero-assist is charged only its Δv; launch losses are a band, not a trajectory. The companion technical report quantifies each of these and names the remedy. For sizing a propulsion system they are the right approximations; for flying, they are the first guess.

Appendix A

Symbols

μ	gravitational parameter GM [km^3/s^2] (only meaning)
$\mathcal{E}$	specific mechanical energy $v^2/2 - \mu/r = -\mu/(2a)$
$\boldsymbol{h}$, h	specific angular momentum $\boldsymbol{r} \times \boldsymbol{v}$
a, e, p	semi-major axis, eccentricity, semi-latus rectum h^2/μ
r_p, r_a	periapsis, apoapsis radius
ν, E, M, F	true, eccentric, mean, hyperbolic anomaly
n, T	mean motion $\sqrt{\mu/a^3}$, period
i, Ω, ω	inclination, right ascension of the ascending node, argument of periapsis
$\Delta\theta$	transfer angle between $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$
f, g	Lagrange coefficients; z, χ universal variable and anomaly
v_c, v_{esc}	circular and escape speed at a radius
$v_{t,p}$, $v_{t,a}$, a_t, t_H	Hohmann transfer periapsis/apoapsis speeds, semi-major axis, flight time
v_∞, $C_3 = v_\infty^2$	hyperbolic excess speed (relative to the planet), launch energy
δ, β, Δ	flyby turning angle; periapsis-to-asymptote angle; aiming radius
r_{SOI}	sphere-of-influence radius $a(m/M)^{2/5}$
λ_1, γ_1, ϕ	lunar arrival angle; angle at Earth; flight-path angle (above local horizontal)
ϕ_L, Az	launch-site latitude, launch azimuth from north
β (Chapter 8)	ballistic coefficient $m/(C_D A)$ [kg/m^2]; H, ρ_p scale height and periapsis density
ζ, σ_i, π_i, R_i	propellant fraction m_p/m_0; structural coefficient $m_s/(m_s + m_p)$; payload ratio $m_{0,i+1}/m_{0,i}$; stage mass ratio $e^{\Delta v_i/c_i}$
c, I_{sp}, g_0	effective exhaust velocity $g_0 I_{sp}$, specific impulse, 9.806 65 m/s^2

Appendix B

Answers to exercises

1.1
$v_c = 3.315\,\mathrm{km/s}$, $T = 123.1\,\mathrm{min}$, $v_{esc} = 4.689\,\mathrm{km/s}$.

1.2
Perigee speed 7.726+3.5 = 11.23 km/s > v_{esc} = 10.926 km/s: unbound. $v_\infty = \sqrt{v^2 - 2\mu/r} = 2.577\,\mathrm{km/s}$, $C_3 = 6.64\,\mathrm{km^2/s^2}$.

1.3
Half the period, 316.5 min: Kepler's equation is odd about $E = \pi$, so $M = \pi$ exactly at apoapsis for any e.

2.1
$\Delta v_1 = 3.082\,\mathrm{km/s}$, $\Delta v_2 = 0.829\,\mathrm{km/s}$ ($v_{t,a} = 0.190$ vs. $v_c = 1.018\,\mathrm{km/s}$), $t_H = 4.98\,\mathrm{days}$. Δv_1 is the TLI of Chapter 6 (3.135 km/s from 185 km); Δv_2 is what the Moon's own gravity makes unnecessary—the Moon "circularizes" the spacecraft for the price of LOI.

2.2
$\Delta v_2 = v_{c,2} - v_{t,a}$; both terms fall with r_2, but $v_{t,a} \to 0$ faster than $v_{c,2} \propto r_2^{-1/2}$ for small ratios and then both vanish; the difference peaks near $r_2/r_1 \approx 5.9$.

2.3
Separately: 3.893+1.514= 5.41 km/s (plane change at GEO) or +3.803 at LEO; combined optimum 4.231 km/s, with almost all of the angle taken at the slow apogee burn.

3.1
$v_\infty = 5.643\,\mathrm{km/s}$; circular 1000 km: hyperbolic speed 59.395 vs. circular 41.809 km/s, $\Delta v = 17.586\,\mathrm{km/s}$ (hopeless); ellipse to 2×10^6 km: 1.277 km/s. Circular optimum at an altitude of 7886×10^3 km.

3.2
$e = 1.489$, $\delta = 84.4°$, $|\Delta v_{helio}| = 6.717\,\mathrm{km/s}$.

3.3
In the planet frame the encounter is a hyperbola with constant energy, so $|\boldsymbol{v_\infty}|$ in equals out; the Sun-frame velocity adds the planet's velocity to a rotated $\boldsymbol{v_\infty}$, and the magnitude of a sum depends on the angle between the vectors.

5.1
$C_3 = 11.87\,\mathrm{km^2/s^2}$, $v_\infty^{arr} = 3.046\,\mathrm{km/s}$, transfer angle 172° (type I), 260 days.

5.2
For a given geometry two ellipses of the same $a > a_{min}$ connect the points (short and long arc), so $\Delta t(a)$ is double-valued; z parametrizes the arc itself and is single-valued.

5.3
Short flights need hyperbolic transfers whose v_∞—and hence C_3 and Δv—grow without bound as $\Delta t \to 0$; the contours bunch into a wall.

6.1
$\lambda_1 = 29.1°$, 54.3 h to perilune, TLI 3.207 km/s, LOI 1.100 km/s.

6.2
A faster transfer arrives at the Moon with a larger geocentric speed, hence a larger v_∞ relative to the Moon; LOI grows with v_∞ (Eq. (3.4)) while TLI grows only with v_0 at perigee, where the Oberth effect makes energy cheap.

Appendix C

Further reading

Bate, Mueller & White, *Fundamentals of Astrodynamics* (Dover): the clearest first book, and the source of the lunar method of Chapter 6. Curtis, *Orbital Mechanics for Engineering Students*: algorithms with worked numbers, the reference for the universal-variable and Lambert code. Vallado, *Fundamentals of Astrodynamics and Applications*: the practitioner's handbook. Sutton & Biblarz, *Rocket Propulsion Elements*, Ch. 4: the propulsion side of the same story. The companion technical report in `docs/report/` documents the implementation, validation and limitations of the package in full.